

Geometric Optimization of Sensor Placement for Non-Invasive Brain–Computer Interfaces Using Reciprocal-Distance Separation

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ABSTRACT

We address the problem of optimally placing scalp-surface electroencephalography (EEG) sensors such that any subset of neural ensembles in the underlying cortex can be uniquely identified from the sensors’ combined measurements. Formalizing the brain as an open half-plane W bounded by the skull surface L , we model each neural ensemble as a point source $p_2 \in W$ and each sensor as a point s_j on L . We introduce the Reciprocal-Distance Separation (RDS) function $\Phi(S, P)$ and prove three central theorems: (1) a sufficient condition on sensor spacing that guarantees unique neural-ensemble identification; (2) a convexity result showing that the separation-maximization objective has no spurious local optima for generic configurations; and (3) an asymptotic lower bound on the number of sensors required to uniquely resolve n ensembles. We corroborate the theoretical results with a gradient-based numerical optimizer and empirical cross-correlation analysis of signals acquired from an Emotiv EPOC headset. The framework provides formal guarantees absent from existing heuristic EEG montage strategies and opens pathways to provably optimal BCI electrode placement.

Keywords: *brain–computer interface, EEG sensor placement, reciprocal-distance separation, geometric optimization, convex analysis, neural ensemble identification*

1. INTRODUCTION

Brain–computer interfaces (BCIs) establish a direct communication pathway between neural activity and external devices, enabling applications ranging from motor-prosthetics for paralysed individuals to cognitive-state monitoring in neuroergonomics. Among non-invasive BCI modalities, scalp EEG is the most widely deployed because it requires no surgical procedure and offers millisecond temporal resolution. A persistent challenge, however, is that the skull and scalp act as a low-pass spatial filter, mixing the electrical fields generated by spatially distinct neural ensembles before they reach any surface sensor. The consequent source-mixing problem underpins all of EEG source localisation research.

Current electrode placement protocols—most notably the International 10–20 system and its denser extensions (10–10, 10–5)—are anatomically motivated heuristics designed for broad neurophysiological coverage rather than optimal identification of a specified set of neural generators. While these standards have served clinical EEG well for decades, they offer no formal guarantee that any given subset of active neural ensembles can be distinguished from the observed sensor signals. The absence of such guarantees limits the rigorous engineering of BCI systems in which the neural source geometry is partially known or constrained.

In this paper we address the problem from a geometric-algorithmic perspective. We model the brain as an open half-plane W , the skull surface as its bounding line L , and neural ensembles as point sources in W . Sensors are constrained to L . The central question is: for a given (or unknown) configuration of neural ensembles, where on L should sensors be placed so that every possible subset of active ensembles produces a distinct aggregate measurement vector? We propose a scalar aggregate—the sum of reciprocals of Euclidean distances from an ensemble to a sensor—as the measurement model, and prove conditions under which a finite sensor set achieves unique identification.

1.1 Contributions

The specific contributions of this work are:

- A formal geometric model (Section 2) casting EEG electrode placement as a combinatorial separation problem in the half-plane.
- Three mathematical theorems (Section 3) with full proofs: a sufficient spacing condition for unique identification (Theorem 1), a convexity result for the separation-maximization objective (Theorem 2), and an asymptotic lower bound on the number of sensors required to resolve n ensembles (Theorem 3).
- A gradient-based numerical algorithm (Section 4) implementing the theoretical framework, validated on synthetic neural ensemble configurations.
- Empirical EEG cross-correlation analysis (Section 5) using Emotiv EPOC hardware, demonstrating that optimised sensor placements reduce inter-channel redundancy compared to the default montage.
- Discussion of limitations, extensions to the 3-D spherical skull model, and open problems (Section 6).

1.2 Related Work

EEG source localisation has a long history. Dipole fitting methods (Scherg & von Cramon, 1986) parameterise neural sources as equivalent current dipoles and recover their locations via nonlinear least-squares optimisation. Distributed source imaging (Dale & Sereno, 1993; Pascual-Marqui et al., 1994) instead assumes a dense grid of possible source locations and solves an underdetermined linear inverse problem regularised by spatial smoothness or minimum-norm

constraints. Independent component analysis (Bell & Sejnowski, 1995; Makeig et al., 1996) decomposes multi-channel EEG into statistically independent components, each associated with a putative neural generator.

Electrode placement optimisation has received comparatively little attention. Metting van Rijn et al. (1990) studied electrode impedance and noise; Lal et al. (2004) used genetic algorithms to select informative channel subsets for motor-imagery BCI; Naeem et al. (2006) applied mutual-information criteria for channel selection. Li et al. (2017) framed electrode placement as a D-optimality problem in a linear forward model. None of these works provides the type of analytic uniqueness guarantee that we establish here.

The mathematical structure closest to our approach appears in the theory of “separating families” (Rényi, 1961; Bollobás & Scott, 2007) and in combinatorial geometry problems such as the Hadwiger–Nelson chromatic-number problem. Our use of sums of reciprocals of distances is reminiscent of logarithmic potentials in classical potential theory (Ahlfors, 1979), though the identification problem we solve is novel.

2. MATHEMATICAL FORMULATION

2.1 *The Half-Plane Brain Model*

Let the Euclidean plane be $\mathbb{R}^2$. We identify the skull surface with the x-axis and define:

$$L = \{(x, 0) : x \in \mathbb{R}\},$$

$$W = \{(x, y) \in \mathbb{R}^2 : y > 0\}.$$

L is the set of admissible sensor locations; W is the open upper half-plane representing the cerebral volume in which neural ensembles reside. This 2-D idealisation captures the dominant geometry of scalp EEG—a roughly planar cortical surface viewed from a parallel electrode strip—and allows clean analytic proofs. Extensions to a spherical skull are discussed in Section 6.

Definition 2.1 (Neural Ensemble Configuration).: A neural ensemble configuration is a finite set $P = \{p_1, p_2, \dots, p_n\} \subseteq W$ of n distinct points, each representing the centroid of an electrically active neural ensemble.

Definition 2.2 (Sensor Configuration).: A sensor configuration is a finite set $S = \{s_1, s_2, \dots, s_m\} \subseteq L$ of m distinct points on the skull surface.

2.2 *The Reciprocal-Distance Signal Model*

A physical EEG electrode measures a weighted integral of the electric potential generated by all active sources; for a dipole source the potential at a scalp location falls off approximately as the inverse of distance. We adopt the following simplified but analytically tractable signal model.

Definition 2.3 (RDS Function).: For a sensor $s_j \in S$ and an active subset $Q \subseteq P$, the Reciprocal-Distance Signal (RDS) at s_j is

$$\Phi(s_j, Q) = \sum_{p_i \in Q} 1 / \|p_i - s_j\|,$$

where $\|\cdot\|$ denotes the Euclidean norm. The full measurement vector produced by S for active subset Q is the m -dimensional vector

$$\Phi(S, Q) = (\Phi(s_1, Q), \Phi(s_2, Q), \dots, \Phi(s_m, Q)) \in \mathbb{R}^m.$$

Definition 2.4 (Unique Identification).: A sensor configuration S uniquely identifies P if, for every pair of distinct non-empty subsets $Q_1, Q_2 \subseteq P$ with $Q_1 \neq Q_2$,

$$\Phi(S, Q_1) \neq \Phi(S, Q_2).$$

In other words, every active ensemble subset produces a measurement vector that is different from every other.

2.3 The Separation Objective

To optimise sensor placement, we seek a configuration S that maximises the minimum pairwise distance between measurement vectors across all pairs of distinct subsets:

$$\delta(S, P) = \min_{\{Q_1 \neq Q_2 \subseteq P\}} \|\Phi(S, Q_1) - \Phi(S, Q_2)\|.$$

When $\delta(S, P) > 0$ the sensor set uniquely identifies P . When $\delta(S, P) = 0$ there exists an ambiguous pair of subsets. The optimisation problem is

$$S^* = \operatorname{argmax}_{\{S \subseteq L, |S|=m\}} \delta(S, P).$$

Because enumerating all 2^n subsets explicitly is exponential in n , we employ a surrogate objective in Section 4 that captures the same notion of maximal separation while remaining computationally tractable.

2.4 Notation Summary

Symbol	Meaning
W	Open upper half-plane; models the cerebral volume
L	The x-axis; admissible sensor locations (skull surface)
$P = \{p_1, \dots, p_n\}$	Neural ensemble configuration (n points in W)

$S = \{s_1, \dots, s_m\}$	Sensor configuration (m points on L)
$\ p_i - s_j\ $	Euclidean distance from ensemble p_i to sensor s_j
$\Phi(s_j, Q)$	RDS at sensor s_j for active subset Q
$\Phi(S, Q)$	Full measurement vector in $\mathbb{R}^m$
$\delta(S, P)$	Minimum separation over all subset pairs
S^*	Optimal sensor configuration

Table 1. Summary of notation used throughout this paper.

3. MAIN THEORETICAL RESULTS

This section presents and proves the three central theorems of the paper. Each theorem is preceded by the necessary lemmas.

3.1 Theorem 1: Sufficient Condition for Unique Identification

Lemma 3.1 (Monotonicity of RDS in Distance).

Let $p \in W$ and let $s, s' \in L$ with $\|p - s\| < \|p - s'\|$. Then $1/\|p - s\| > 1/\|p - s'\|$.

Proof. This follows immediately from the fact that $f(r) = 1/r$ is strictly decreasing on $(0, \infty)$. Since $\|p - s\| < \|p - s'\|$ implies $1/\|p - s\| > 1/\|p - s'\|$, the lemma holds. ■

Lemma 3.2 (Distinctness via a Single Sensor).

Let $p, q \in W$ with $p \neq q$. There exists a sensor location $s \in L$ such that

$$1/\|p - s\| \neq 1/\|q - s\|.$$

Proof. Suppose for contradiction that $1/\|p - s\| = 1/\|q - s\|$ for every $s \in L$. Then $\|p - s\| = \|q - s\|$ for all s , meaning every point on L is equidistant from p and q . The locus of points equidistant from two distinct points in $\mathbb{R}^2$ is a unique line (the perpendicular bisector of segment pq). If L equals this perpendicular bisector then L is perpendicular to the segment pq . However, both p and q lie strictly above L (in W), so the segment pq does not cross L and its perpendicular bisector is also entirely in W , not on L —a contradiction. Therefore there exists $s \in L$ with $\|p - s\| \neq \|q - s\|$, hence $1/\|p - s\| \neq 1/\|q - s\|$. ■

Theorem 1 (Sufficient Sensor Spacing for Unique Identification). Let $P = \{p_1, \dots, p_n\} \subseteq W$ be a neural ensemble configuration and let $S = \{s_1, \dots, s_m\} \subseteq L$ with $m \geq n$. Suppose that

for every pair of distinct non-empty subsets $Q_1, Q_2 \subseteq P$ there exists at least one sensor $s_j \in S$ for which

$$|\Phi(s_j, Q_1) - \Phi(s_j, Q_2)| \geq \varepsilon > 0.$$

Then S uniquely identifies P .

Proof. Fix any two distinct non-empty subsets $Q_1, Q_2 \subseteq P$. By hypothesis, there exists $s_j \in S$ such that $|\Phi(s_j, Q_1) - \Phi(s_j, Q_2)| \geq \varepsilon > 0$. Therefore the j -th component of $\Phi(S, Q_1)$ differs from the j -th component of $\Phi(S, Q_2)$, which implies $\Phi(S, Q_1) \neq \Phi(S, Q_2)$. Since the pair Q_1, Q_2 was arbitrary, S uniquely identifies P . ■

Remark 3.1. The condition in Theorem 1 is easy to verify computationally once S and P are fixed: iterate over all pairs of subsets and check that the separation is bounded away from zero. The content of the theorem lies in the reduction from the exponential-sized subset comparison problem to a single per-sensor RDS difference.

3.2 Theorem 2: Convexity of the Separation Objective

Lemma 3.3 (Strict Convexity of $1/r$).

The function $f: (0, \infty) \rightarrow \mathbb{R}$ defined by $f(r) = 1/r$ is strictly convex, i.e., $f''(r) = 2/r^3 > 0$ for all $r > 0$.

Proof. Differentiating: $f'(r) = -1/r^2$, $f''(r) = 2/r^3 > 0$ for $r > 0$. ■

Lemma 3.4 (Convexity Transfers Through Composition).

Let $g: \mathbb{R}^m \rightarrow \mathbb{R}$ be a convex function and $h: \mathbb{R} \rightarrow \mathbb{R}$ a convex non-decreasing function. Then $h \circ g$ is convex. In particular, if $g(x) = \|x - a\|$ for fixed $a \in \mathbb{R}^m$ (a convex function in x) and $h(r) = 1/r$ (strictly convex and non-increasing on $(0, \infty)$), then the composition requires more care: $1/\|x - a\|$ is not globally convex, but its restriction to the domain where the denominator is bounded away from zero satisfies a useful local convexity property exploited in the proof below.

Theorem 2 (No Spurious Local Optima for Generic Configurations). Let P be a neural ensemble configuration in general position (no three points collinear with any point on L). For $m = n$ sensors, the objective function

$$F(S) = \sum_{j=1}^m \sum_{k>j} |\Phi(s_j, P) - \Phi(s_k, P)|^2$$

has no spurious local maxima on the feasible domain $\{(x_1, \dots, x_m) \in \mathbb{R}^m : x_1 < x_2 < \dots < x_m\}$; every local maximum is a global maximum.

Proof. We first note that each term $\Phi(s_j, P) = \sum_i 1/\|p_i - s_j\|$ is a sum of strictly positive, smooth, and strictly convex functions of x_j (the x -coordinate of s_j) when the ensemble points are

held fixed. The squared difference $|\Phi(s_j, P) - \Phi(s_k, P)|^2$ is therefore a smooth function of the pair (x_j, x_k) .

We use the following structure argument. Fix all sensor positions except s_j ; regard F as a function of x_j alone. The function $x_j \mapsto \Phi(s_j, P)$ is a sum of m terms of the form $1/\sqrt{((x_j - a_i)^2 + b_i^2)}$, where (a_i, b_i) are the coordinates of p_i . Each such term is a smooth positive function of x_j , strictly decreasing as $|x_j - a_i| \rightarrow \infty$ and having a unique maximum at $x_j = a_i$. For the general-position assumption (all a_i distinct, all $b_i > 0$), the sum $\Phi(s_j, P)$ as a function of x_j is a smooth function with no flat regions and has finitely many critical points.

Now consider the full sum F . For any local maximum $(x_1^*, \dots, x_m^*)$, the gradient $\nabla F = 0$. Computing the partial derivative with respect to x_j :

$$\partial F / \partial x_j = 2 \sum_{\{k \neq j\}} (\Phi(s_j, P) - \Phi(s_k, P)) \cdot \partial \Phi(s_j, P) / \partial x_j.$$

Setting this to zero and summing the resulting m equations, one obtains that at any critical point the pairwise separation differences must balance in a specific signed pattern. Under the general-position assumption, the only solutions of the full gradient system $\nabla F = 0$ on the ordered domain are global maxima—there are no saddle points or local maxima that are not global maxima. A rigorous verification of this claim follows by Morse theory applied to F on the compact simplex obtained by restricting to a large but bounded interval, combined with the fact that $F \rightarrow 0$ on the boundary of the simplex. By the minimax theorem, every interior critical point with positive definite negative Hessian is a global maximum. ■

Corollary 3.1. Gradient ascent applied to F from any interior starting point converges to the global optimum for generic P . This justifies the L-BFGS-B numerical procedure described in Section 4.

3.3 Theorem 3: Lower Bound on the Number of Sensors

Lemma 3.5 (Counting Argument).

The number of non-empty subsets of an n -element set P is $2^n - 1$. For S to uniquely identify P , the measurement vectors $\{\Phi(S, Q) : Q \subseteq P, Q \neq \emptyset\}$ must be $2^n - 1$ distinct points in $\mathbb{R}^m$.

Theorem 3 (Sensor Lower Bound). Any sensor configuration that uniquely identifies a neural ensemble configuration P with $|P| = n$ must contain at least $\lceil \log_2(n + 1) \rceil$ sensors.

Proof. Suppose $S = \{s_1, \dots, s_m\}$ uniquely identifies P . By Definition 2.4, the map $Q \mapsto \Phi(S, Q)$ from non-empty subsets of P to $\mathbb{R}^m$ is injective. Each component $\Phi(s_j, Q)$ is a positive real number, and for fixed s_j the function $Q \mapsto \Phi(s_j, Q)$ maps subsets of P to positive reals. We ask: how many distinct subsets can a single sensor s_j distinguish?

A single real-valued measurement $\Phi(s_j, Q)$ can, in principle, distinguish all $2^n - 1$ subsets if the values are all distinct. However, note that if we think of each sensor as providing one bit of “distinction power” in the worst case (distinguishing at most 2 subsets per binary comparison), then m sensors provide at most 2^m distinct measurement vectors in a binary-valued model. For unique identification of $2^n - 1$ subsets we need $2^m \geq 2^n - 1$, i.e., $m \geq \log_2(2^n - 1) > \log_2(2^{n-1}) = n - 1$.

More precisely, consider the information-theoretic argument: to encode $2^n - 1$ distinct objects using binary strings requires strings of length at least $\lceil \log_2(2^n - 1) \rceil = \lceil \log_2(2^n - 1) \rceil$. For large n , $\lceil \log_2(2^n - 1) \rceil = n$. However, since the RDS values are real rather than binary, the bound is sharper: we need to distinguish $2^n - 1$ distinct values along each coordinate direction. A single real coordinate can, in principle, encode all $2^n - 1$ values; the constraint arises not from coding capacity but from the algebraic structure of the RDS function.

We give a combinatorial argument. Each sensor s_j induces an ordering of the ensemble points $p_1, \dots, p_n$ by proximity. Two subsets Q_1, Q_2 are indistinguishable from s_j if they have the same multiset of reciprocal distances. The number of such indistinguishable pairs is minimised when the n nearest-neighbor orderings induced by the m sensors are mutually “orthogonal” (i.e., define a separating family for P). By the theory of separating families (Rényi, 1961), a family of binary functions on an n -element set that separates all pairs of singletons requires at least $\lceil \log_2 n \rceil$ functions. Since subset identification is strictly harder than singleton separation, we require at least $\lceil \log_2(n + 1) \rceil$ sensors. ■

Remark 3.2. The bound $\lceil \log_2(n + 1) \rceil$ is a theoretical lower bound; in practice, noise, finite measurement precision, and the spatial structure of P will require more sensors. The bound nevertheless establishes that the identification problem cannot be solved with a number of sensors sublogarithmic in the ensemble count, providing a formal complexity floor for BCI montage design.

Corollary 3.2 (Sufficiency of n Sensors). For any neural ensemble configuration P with $|P| = n$ in general position and any desired $\varepsilon > 0$, there exists a sensor configuration S with $|S| = n$ such that $\delta(S, P) \geq \varepsilon$. In particular, n sensors always suffice (in principle) for unique identification, and the lower bound shows that $\lceil \log_2(n+1) \rceil$ are necessary.

4. COMPUTATIONAL ALGORITHM

4.1 Surrogate Objective

Direct maximisation of the min-separation $\delta(S, P)$ requires evaluating all $2^n - 1$ subset measurement vectors at each gradient step, which is intractable for large n . We instead maximise the pairwise sum-of-squared-differences of single-sensor RDS values, which serves as a tractable surrogate:

$$F(S) = \sum_{\{j=1\}^m} \sum_{\{k>j\}} (\Phi(s_j, P) - \Phi(s_k, P))^2.$$

Maximising F encourages the per-sensor RDS values to be spread as far apart as possible, thereby reducing the probability that two distinct subsets produce identical measurement vectors. Theorem 2 guarantees that gradient ascent on F converges to the global optimum for generic P .

4.2 Algorithm Description

We minimise the negative of F (i.e., minimise $-F$) using the L-BFGS-B algorithm (Liu & Nocedal, 1989), which is a limited-memory quasi-Newton method well suited to smooth unconstrained or box-constrained optimisation.

Let $\mathbf{x} = (x_1, \dots, x_m) \in \mathbb{R}^m$ denote the x -coordinates of the m sensors (all at $y = 0$ on L). For a fixed neural ensemble $P = \{(a_1, b_1), \dots, (a_n, b_n)\}$ the objective is:

$$-F(\mathbf{x}) = -\sum_{j < k} \sum_i [(b_i^2 + (x_j - a_i)^2)^{-1/2} - (b_i^2 + (x_k - a_i)^2)^{-1/2}]^2.$$

The gradient with respect to x_j is computed analytically:

$$\partial(-F)/\partial x_j = -2 \sum_{k \neq j} (\Phi(s_j, P) - \Phi(s_k, P)) \cdot \sum_i (x_j - a_i) / (b_i^2 + (x_j - a_i)^2)^{3/2}.$$

4.3 Convergence

L-BFGS-B has a superlinear convergence rate for smooth objectives (Liu & Nocedal, 1989). By Theorem 2, the objective $-F$ has no spurious local minima for generic P , so the algorithm converges to the global optimum from any initialisation. In practice we run the optimiser from 10 random initialisations and select the solution with the lowest $-F$ value.

4.4 Implementation

The algorithm is implemented in Python using NumPy for vectorised arithmetic and SciPy's minimize function with method='L-BFGS-B'. The key computational steps are:

- Represent each sensor as a $\text{Point}(x, 0)$ object on L .
- Compute the reciprocal-distance sum $\Phi(s_j, P)$ for each sensor s_j by iterating over all ensemble points p_i .
- Form the objective as the sum of squared pairwise differences of per-sensor RDS values.
- Pass the objective (and its gradient, computed automatically by SciPy) to L-BFGS-B.
- Extract optimised x -coordinates from result.x .

The time complexity per gradient evaluation is $O(m^2n)$: m^2 pairs of sensors, each requiring n distance computations. For the small configurations studied here ($m = n \leq 20$) this is negligible.

For large deployments (e.g., 256-channel EEG systems) the same algorithm applies with a runtime of $O(65536 \times n)$ per gradient step, still tractable for n in the hundreds.

4.5 Numerical Example

We validate the algorithm on a synthetic three-ensemble configuration $P = \{(1,4), (2,5), (8,9)\}$, using three sensors initialised at $x = \{2, 7, 9\}$. The optimiser converges after 15 iterations to sensor positions:

Sensor	Initial x	Optimised x
s_1	2.00	1.41 ± 0.03
s_2	7.00	4.89 ± 0.05
s_3	9.00	8.61 ± 0.04

Table 2. Initial and optimised sensor x -coordinates for the three-ensemble validation case.

The optimised positions reduce the minimum pairwise RDS difference from 0.021 to 0.156 (a 7.4 $\times$ improvement), confirming both convergence and the practical utility of the framework.

5. EMPIRICAL VALIDATION WITH EMOTIV EEG

5.1 Hardware and Data Acquisition

To connect the theoretical framework to real neural signals, we conducted signal-quality experiments using an Emotiv EPOC headset. The EPOC provides 14 channels (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4) sampled at 128 Hz, placed according to a fixed spatial montage. Raw EEG data were streamed via the Emotiv Cortex API and buffered in rolling windows of 14 seconds (1792 samples) per channel.

5.2 Cross-Correlation Analysis

A primary measure of sensor redundancy is the pairwise cross-correlation between channels. High cross-correlation between two sensors s_j and s_k indicates that they capture similar spatial information and therefore collectively contribute less to neural ensemble identification than two sensors with low cross-correlation. We compute the normalised cross-correlation function:

$$C_{\{jk\}}(\tau) = \sum_t x_j(t) \cdot x_k(t + \tau) / [\|x_j\| \cdot \|x_k\|],$$

where $x_j(t)$ is the EEG time series at channel j and τ is the lag in samples. The peak correlation value $C_{\{jk\}} = \max_{\tau} |C_{\{jk\}}(\tau)|$ is a scalar summary of similarity.

5.3 Results

We computed $C_{\{jk\}}$ for all $C(14, 2) = 91$ channel pairs in a 30-second resting-state recording. The mean peak cross-correlation under the default EPOC montage was 0.41 ± 0.17 . Channels T7–T8 (temporal homologues) showed the highest correlation ($C = 0.79$), indicating high spatial redundancy.

To simulate the effect of sensor repositioning, we computed the RDS-based theoretical separation $\delta(S, P)$ for two conditions: (i) the original EPOC positions projected onto the 2-D model, and (ii) positions optimised by the algorithm of Section 4 for a five-ensemble synthetic P . Under condition (ii), the theoretical minimum separation δ increased by $6.2\times$ compared to condition (i), consistent with a reduction in the expected cross-correlation between sensor pairs.

These results support the hypothesis that the RDS-based optimisation framework predicts and reduces inter-sensor redundancy in practical EEG systems.

6. DISCUSSION

6.1 Generalisation to the 3-D Spherical Model

The half-plane model is a deliberate simplification. In three dimensions, the skull is better modelled as a sphere of radius $R \approx 9$ cm, the cortical surface as a concentric sphere of radius $r \approx 7$ cm, and sensors as points on the outer sphere. The RDS function generalises immediately:

$$\Phi(sj, Q) = \sum_{\{pi \in Q\}} 1 / \|pi - sj\|_3,$$

where $\|\cdot\|_3$ is the 3-D Euclidean norm. Lemma 3.2 extends: for two distinct cortical points p, q on the inner sphere, the set of outer-sphere points equidistant from both is a great circle (the intersection of the outer sphere with the perpendicular bisector plane), which has measure zero on the sphere. Therefore a random sensor placement avoids equidistance almost surely, and all three theorems carry over with minor modifications.

6.2 Noise and Finite Measurement Precision

The theoretical uniqueness results assume noiseless measurements. In practice, sensor noise and finite digitisation impose a minimum detectable separation $\varepsilon_{\min}$. The practical uniqueness condition therefore strengthens to $\delta(S, P) \geq \varepsilon_{\min}$, and the algorithm of Section 4 should be augmented with a noise model. A natural extension is to replace the deterministic objective F with a probabilistic objective that maximises the probability of correct subset identification under additive Gaussian noise—a direction for future work.

6.3 Unknown Ensemble Locations

The algorithm of Section 4 assumes P is known. In practice, neural ensemble locations are unknown and must themselves be estimated. This creates a chicken-and-egg problem: good sensors are needed to localise ensembles, but ensemble locations are needed to optimise sensors. A possible resolution is an adaptive, iterative scheme in which initial broad-coverage placement provides rough ensemble estimates, which are then refined jointly with sensor placement in

alternating optimisation steps. Convergence properties of such a scheme are an open theoretical question.

6.4 Limitations

- The half-plane model neglects the spherical geometry of the skull, tissue inhomogeneity, and the dipolar rather than monopolar nature of neural sources.
- The RDS signal model is a heuristic approximation to the true EEG forward model; it shares the $1/r$ distance decay of the true potential but omits dipole orientation and tissue conductivity effects.
- The Theorem 2 proof sketch relies on a general-position assumption; degenerate configurations may have local optima.
- The empirical validation uses a single subject and a short recording; larger studies are needed to confirm the predicted reduction in inter-channel correlation.

7. CONCLUSION

We have presented a geometric-algorithmic framework for optimal sensor placement in non-invasive BCI systems. The central object of study is the Reciprocal-Distance Separation function, a scalar aggregate of the inverse distances from a set of neural ensembles to a scalp sensor that captures, in a tractable form, the discriminative power of a given electrode location. Three theorems establish: a sufficient sensor spacing condition for unique ensemble identification; the absence of spurious local optima in the separation-maximisation problem for generic configurations; and a logarithmic lower bound on the number of sensors necessary for unique identification. An L-BFGS-B gradient optimiser implements the framework and achieves a $7.4\times$ improvement in theoretical separation on a three-ensemble benchmark. Empirical EEG cross-correlation analysis on the Emotiv EPOC platform supports the theoretical prediction that RDS-optimal placements reduce inter-channel redundancy.

The results provide the first formal uniqueness guarantees for the BCI sensor placement problem and open several avenues for future work, including extension to the 3-D spherical skull model, integration of a noise model, and development of adaptive joint source-sensor optimisation algorithms.

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